3.2.9 Originally, classification systems were based on observable features but more recent approaches draw on a wider range of evidence to clarify relationships between organisms.

Genetic comparisons	Genetic comparisons can be made between different species by direct examination of their DNA or of the proteins encoded by this DNA.
DNA	Comparison of DNA base sequences is used to elucidate relationships between organisms. These comparisons have led to new classification systems in plants. Similarities in DNA may be determined by DNA hybridisation.
Proteins	Comparisons of amino acid sequences in specific proteins can be used to elucidate relationships between organisms. Immunological comparisons may be used to compare variations in specific proteins. **Candidates should be able to** interpret data relating to similarities and differences in base sequences in DNA and in amino acid sequences in proteins to suggest relationships between different organisms.
Behaviour	Courtship behaviour as a necessary precursor to successful mating. The role of courtship in species recognition.

3.2.10 Adaptation and selection are major components of evolution and make a significant contribution to the diversity of living organisms.	
Antibiotics	Antibiotics may be used to treat bacterial disease. One way in which antibiotics function is by preventing the formation of bacterial cell walls, resulting in osmoticlysis.
Genetic variation in bacteria	DNA is the genetic material in bacteria as well as in most other organisms. Mutations are changes in DNA and result in different characteristics. Mutations in bacteria may result in resistance to antibiotics. Resistance to antibiotics may be passed to subsequent generations by vertical gene transmission. Resistance may also be passed from one species to another when DNA is transferred during conjugation. This is horizontal gene transmission. Antibiotic resistance in terms of the difficulty of treating tuberculosis and MRSA. **Candidates should be able to** • apply the concepts of adaptation and selection to other examples • evaluate methodology, evidence and data relating to antibiotic resistance • discuss ethical issues associated with the use of antibiotics • discuss the ways in which society uses scientific knowledge relating to antibiotic resistance to inform decision-making.

3.2.11 Biodiversity may be measured within a habitat.	
Species diversity	Diversity may relate to the number of species present in a community. The influence of deforestation and the impact of agriculture on species diversity.
Index of diversity	An index of diversity describes the relationship between the number of species and the number of individuals in a community. Calculation of an index of diversity from the formula $d = \dfrac{N\ (N-1)}{\Sigma\ n\ (n-1)}$ where N = total number of organisms of all species and n = total number of organisms of each species **Candidates should be able to** • calculate the index of diversity from suitable data • interpret data relating to the effects of human activity on species diversity and be able to evaluate associated benefits and risks • discuss the ways in which society uses science to inform the making of decisions relating to biodiversity.

Biological principles

When they have completed this unit, candidates will be expected to have an understanding of the following principles.

- A species may be defined in terms of observable similarities and the ability to produce fertile offspring.
- Living organisms vary and this variation is influenced by genetic and environmental factors.
- The biochemical basis and cellular organisation of life is similar for all organisms.
- Genes are sections of DNA that contain coded information as a specific sequence of bases.
- During mitosis, the parent cell divides to produce genetically identical daughter cells.
- The relationship between size and surface area to volume ratio is of fundamental importance in exchange.

An understanding of these principles may be required in the A2 Units where it may contribute to the assessment of synoptic skills.

Investigative and practical skills

Opportunities to carry out the necessary practical work should be provided in the context of this unit. Candidates will be assessed on their understanding of *How science works* in all three AS Units. To meet this requirement, candidates are required to

- use their knowledge and understanding to pose scientific questions and define scientific problems
- carry out investigative activities, including appropriate risk management, in a range of contexts
- analyse and interpret data they have collected to provide evidence
- evaluate their methodology, evidence and data, resolving conflicting evidence.

They will require specific knowledge of the following skills and areas of investigation. The practical work undertaken should include investigations which consider

- collection and analysis of data relating to intraspecific variation
- use of an optical microscope to examine temporary mounts of plant cells, tissues or organs
- measurement of the rate of water uptake by means of a simple potometer.